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1. Introduction

The proliferation of electronic devices equipped with wireless communication functions is expected to increase significantly. This growth is driven by the expanding applications of 5G (5th Generation Mobile Communication System), which commenced commercial service in 2020, and by planned information access to cyberspace as envisioned in Society 5.0. Consequently, demand for measuring the phase noise of numerous oscillators that generate wireless communication carrier frequencies is anticipated to increase. Phase noise analyzers are costly, and precise phase noise measurements require a thousand seconds or more. Therefore, cost reduction and the capability for simultaneous multiple measurements are desirable. Measuring phase noise—the phase fluctuations in a periodic signal—requires a phase reference. For this reason, precise phase noise measurement conventionally employs a reference oscillator (REF) in addition to the device under test (DUT) for comparative measurement. This paper proposes an all-digital phase noise measurement method that eliminates the reference oscillator required for phase noise measurement, thereby reducing the circuit scale by half compared to traditional approaches.

2. Principles of phase noise measurement

Precise phase noise measurement is categorized into PLL-based and all-digital types. This chapter describes the principles and characteristics of methods using analog PLLs, the conventional mainstream approach for phase noise measurement, and all-digital methods that do not employ PLLs.

2.1. Phase noise measurement using down conversion by analog PLLs

Fig. 1 shows a block diagram of the phase noise measurement method using analog PLLs. In this method, the phase noise near the oscillation frequency of the DUT is down-converted to the baseband by configuring a PLL using a REF with a frequency variable function, and then low-noise amplifier(LNA) amplifies a small voltage signal proportional to the phase noise $\phi(t)$. after that we analyze it as spectrum analysis. In this method, the obtained phase noise spectrum includes the phase noise from DUT's, the PLL-circuit's and the LNA's. Therefore, we need to prepare two identical PLLs and their subsequent circuits and extract only the DUT's phase noise by performing correlation processing. If there is no correlation between the phase noise in both paths, the uncorrelated noise is reduced by $10 \log M$ dB(M is the number of correlation averages or operations). The analog PLL method requires PLL circuits for frequency down-conversion and LNAs. Furthermore, we need to calibrate for each measurement because the mixer's phase detection sensitivity varies depending on the DUT's frequency and signal power. Additionally, to accommodate various frequencies, REFs

need a wide variable frequency range. Consequently, the circuit design techniques that configuring complex PLLs and the cost of the components are drawbacks.

2.2. All digital phase noise measurement

Fig. 2 shows a block diagram of an all-digital phase noise measurement method. In this method, an Analog-to-Digital Converter (ADC) directly samples the outputs of the Device Under Test (DUT) and the Reference (REF). A Digital Down Converter (DDC) unit then calculates the phase through digital operations. This approach eliminates the need for analog circuits such as Phase-Locked Loop (PLL) circuits and low-noise amplifiers (LNAs). The measurement's dynamic range is limited by the ADC's quantization noise and the phase noise of the clock (CLK) driving the ADC. Like PLL-based systems, the ADC's phase noise is reduced through correlation processing of Fast Fourier Transform (FFT) outputs obtained from two paths. Furthermore, clock phase noise is reduced by subtracting the REF signal phase from the DUT signal phase after the DDC stage. An advantage of the all-digital method, unlike PLL-based methods, is that the REF does not require a variable frequency capability or to operate at the same frequency as the DUT. This is because the DDC unit, which converts the time waveform to phase, can generate arbitrary frequencies using a Numerically Controlled Oscillator (NCO). Fig. 3 shows a block diagram of the DDC unit. In the DDC unit, the discrete-time waveform sampled by the ADC is multiplied by sine waves of the same frequency but with a 90° phase difference (quadrature signals). The phase is then calculated from the resulting In-phase (I) and Quadrature (Q) components using a $\tan^{-1}(Q/I)$ operation. Therefore, in the all-digital method, unlike analog PLL-based systems, there are no restrictions on the REF frequency. However, if the DUT and REF frequencies differ, residual clock phase noise (ϕ_{CLK}) remains, proportional to the ratio of their frequency difference to the sampling frequency. Therefore, it is desirable for both frequencies to be identical.

3. All Digital Phase Noise Measurement Method without Use of Reference Oscillator

The previous section described phase noise measurement using analog PLL and all-digital methods. The all-digital method is fundamentally based on the conventional analog PLL architecture, replacing the mixer at the signal input with an Analog-to-Digital Converter (ADC). Consequently, this system requires not only two reference signals (REFs) but also four channels for the ADCs and signal processing system. Here, focusing on the paths of channels 2 and 3 in Fig. 2, we observe that both channels share ϕ_{DUT} as a common phase noise component, whereas the other noise sources, ϕ_{ADC} and ϕ_{CLK} , are independent components. This suggests that it is possible to measure ϕ_{DUT} exclusively through correlation processing, even in a configuration that excludes channels 1 and 4. In this

chapter, we propose a measurement technique that eliminates the need for the reference oscillator required by conventional all-digital phase noise measurement systems. We will then verify the level of phase noise that can be measured with this new method.

3.1. All-digital phase noise without a reference oscillator Measurement system configuration

Figure 4 presents a block diagram of the proposed phase noise measurement system, which operates without a reference oscillator. This configuration is derived from the architecture in Fig. 2 by removing two of each of the following components: REFs, ADCs, DDCs, and subtractors. In the signal paths of ADC1 and ADC2, ϕ DUT is the only common phase noise component, while all other noise sources are independent. As a result, correlation processing isolates and measures only the power spectral density of ϕ DUT, denoted as $S_{\phi \text{ DUT}}$. Compared to the conventional all-digital method, this approach halves both the circuit size and the computational load in the DDC. The cost savings are particularly significant, as the DDC stage requires a large-scale FPGA with numerous multipliers and Look-Up Tables (LUTs) to implement high-order digital filters and $\tan^{-1}$ calculations. Furthermore, for a given level of hardware complexity, this architecture makes it possible to measure two DUTs simultaneously.

3.2. Verification of phase noise reduction in clocks

In all-digital phase noise measurement, while the reduction of ADC phase noise through correlation processing has already been established, this paper investigates the residual amount of the sampling clock phase noise (ϕ CLK) after correlation processing. The digitizer board used in this study generates its sampling clock using an onboard PLL IC, based on a signal from a stable frequency source such as a crystal oscillator. As it was difficult to extract the sampling clock from the board to measure ϕ CLK directly, we measured it indirectly using the configuration shown in Fig. 5. By inputting independent oscillator signals into two ADCs driven by a common clock, the only common phase noise component between channels 1 and 2 is that of ϕ CLK, allowing only the clock's phase noise to be measured. Furthermore, by adopting the configuration shown in Fig. 6 to supply the clock to the ADCs, we can measure the residual clock phase noise ($S_{\phi \text{ DUTresidual}}$) that has been reduced by correlation processing. For this measurement, we used an ADC with a full-scale range of 3.8 V and 16-bit resolution (ENOB: 12.1 bit), and the sampling frequency was set to 160 MHz. A 10 MHz crystal oscillator was used as the source for both the input signal (OSC) and the clock generation. Figure 7 shows the measured double-sideband (DSB) phase noise results for the sampling clock and the residual phase noise. The black line in the figure represents the phase noise characteristics of the sampling clock measured with the Fig. 5 configuration, while the colored curves show the results measured with the Fig. 6 configuration. The blue, red, and yellow lines correspond to $M = 10^4$, $M = 10^5$, and $M = 10^6$,

respectively. In the 300 kHz to 1 MHz region, it can be confirmed that for every tenfold increase in M , the phase noise decreases by 5 dB. Moreover, since the fluctuations in the cross-spectrum have not yet converged, it is evident that the phase noise can be reduced further by increasing M .

4. Comparison of phase noise measurement results between the proposed method and the conventional method

4.1. Measurement system configuration for obtaining measurement results using both the proposed method and the conventional method

Figure 8 shows the configuration for comparing the conventional method with the proposed all-digital phase noise measurement method, which does not use a reference oscillator. This setup makes it possible to compare the phase noise characteristics of the two methods on the exact same DUT signal instance, eliminating the need for separate measurements. A 10 MHz DDS frequency synthesizer was used as the DUT (Device Under Test).

4.2. Comparison of phase noise characteristics

Figure 9 shows the phase noise measurement results from the conventional and proposed methods. In the same figure, the phase noise of the clock measured in Fig. 7 is also shown as a gray line. These results indicate that both the conventional and proposed methods yield comparable phase noise characteristics. Although a difference in the spectral variance is observed near the 100 Hz offset frequency, this is because the clock's phase noise is approximately 10 dB higher than that of the DUT. Therefore, it is considered that by further increasing M , the measurement results from the two methods will eventually converge.

Conclusion

In this work, we introduce a new method for all-digital phase noise measurement that eliminates the need for a reference oscillator, replacing the conventional technique. We have demonstrated that phase noise measurement is possible with a circuit half the size of the previous implementation.